Wearable Stress Detection using Deep Learning and Optimization (OptiBME 2026)

Aman Chourasia

Indian Institute of Technology Patna
Bihta, Patna, Bihar 801106, India

March 2026

AMAN
CHOURASIA

Table of Contents

Overview

Dataset

Challenges

Preprocessing

Results

Conclusion

Project Objective

AMAN CHOURASIA

Goal: Develop a deep learning framework for wearable stress classification

Core objectives:

- Build a preprocessing pipeline for WESAD
- Use sliding window segmentation for physiological signals
- Transform windows into spectrograms using STFT
- Train a CNN for baseline, stress, and amusement classification
- Evaluate independent subject performance using LOSO
- Improve generalization through specific subjects and finetuning

Outcome: A subject adaptive stress detection pipeline for wearable sensing

Motivation and Problem Setting

- Physiological signals reflect autonomic nervous system activity
- Wearable devices enable continuous stress monitoring
- Raw signals are noisy, asynchronous, and subject dependent
- A robust model must handle inter subject variability

Dataset and Signals

AMAN CHOURASIA

Dataset: WESAD (Wearable Stress and Affect Detection)

Subjects: 15 subjects

Signals used from the wrist device:

- Electrodermal Activity (EDA)
- Skin Temperature (TEMP)
- Blood Volume Pulse (BVP)

Classes:

- Baseline
- Stress
- Amusement

Dataset Characteristics

- Multimodal physiological recordings from controlled experiments
- Different sampling rates across modalities
- Class imbalance across windows
- Strong subject-to-subject variation in signal patterns

Key Challenges

AMAN CHOURASIA

- Signals recorded at different frequencies
- Labels need alignment with physiological streams
- Continuous series cannot be fed directly to the model
- Large inter-subject variability limits generalization
- Class imbalance affects training stability

Need: Convert raw wearable data into structured learning samples

Signal Alignment

AMAN CHOURASIA

Preprocessing steps:

- Downsampled high-frequency BVP to match lower-rate signals
- Aligned EDA, TEMP, BVP, and label streams temporally
- Truncated all sequences to a common length
- Retained only baseline, stress, and amusement labels

Result? Synchronized multimodal time series suitable for segmentation

Data Cleaning and Normalization

AMAN CHOURASIA

Label cleaning:

- Removed undefined or irrelevant labels
- Kept the three target classes only

Normalization:

- Standardization applied using training statistics
- Mean-centered and variance-scaled features
- Same parameters reused for test data

Purpose: Improve stability and consistency during training

Sliding Window Segmentation

AMAN CHOURASIA

Why segment?

- Continuous signals are transformed into fixed-size samples

Method:

- Window length: 180
- Stride: 60
- Majority voting used to assign each window label

Output:

$$X \in \mathbb{R}^{N \times T \times C}, \quad y \in \mathbb{R}^{N}$$

Time–Frequency Representation via STFT

AMAN CHOURASIA

Transformation:

- Apply Short-Time Fourier Transform to each window
- Convert each modality into a spectrogram
- Use magnitude and log scaling for stable learning

Final model input:

- 3-channel spectrogram
- Channels correspond to EDA, TEMP, and BVP

Benefit: Captures temporal and spectral structure jointly

CNN Architecture

AMAN CHOURASIA

Input: Spectrogram tensor of shape $3 \times F \times T$

Network blocks:

- Convolution
- Batch normalization
- ReLU activation
- Pooling
- Dropout

Classifier:

- Adaptive average pooling
- Fully connected layers
- Output logits for 3 classes

Training Strategy and LOSO Evaluation

AMAN CHOURASIA

Training setup:

- Leave-One-Subject-Out cross-validation
- Weighted cross-entropy loss for class imbalance
- AdamW optimizer
- Cosine annealing learning-rate schedule
- Label smoothing and gradient clipping

Purpose: Measure true cross-subject generalization

Descriptive Analysis of the Dataset

- After segmentation, the dataset contains 2172 windows
- Each sample has shape 180×3
- The final training set remains imbalanced across classes
- Subject-wise signal characteristics are highly heterogeneous

Global Model Performance

LOSO results:

- Mean accuracy: 58.80%
- Macro F1-score: 50.56%
- Precision: 57.41%
- Recall: 57.20%

Interpretation: The global model learns useful patterns, but generalization remains limited

Per-Subject Performance Variation

AMAN CHOURASIA

Observed variability:

- Better performance on some subjects such as S2 and S4
- Lower performance on subjects such as S5 and S17
- Results reflect strong inter-subject differences in physiology

Conclusion: Subject-independent stress classification is challenging

Personalization via Subject-Specific Fine-Tuning

AMAN CHOURASIA

Approach:

- Start from the global LOSO-trained model
- Use about 20% of subject data for fine-tuning
- Freeze feature extractor initially
- Unfreeze selected layers for partial adaptation

Effect: The model adapts to subject-specific signal distributions

Personalized Model Results

AMAN CHOURASIA

Performance after fine-tuning:

- Accuracy improves to approximately 80%
- Macro F1-score improves to approximately 70%
- Gains are obtained using limited subject-specific data

Inference: Personalization is essential for practical wearable deployment

Visual Analysis (in report pdf)

Figure 1, Figure 2, and Figure 3 in the report show the raw signals, spectrograms, and confusion matrices for global and personalized models, respectively. Table 1 and 2, shows model performance metrics and comparisons respectively!

Summary and Future Work

AMAN CHOURASIA

Summary:

- Built a complete wearable stress detection pipeline
- Used spectrograms and a CNN for classification
- Evaluated subject-independent generalization with LOSO
- Showed that personalization improves performance substantially

Future work:

- Larger and more diverse datasets
- Hybrid CNN–RNN or transformer models
- Additional modalities such as respiration or accelerometry
- Real-time deployment and user-level adaptation